13. Fostering self-regulated entrepreneurial learning in entrepreneurship education

Manuel Fahrbach, Tobias Jenert, Alexander Fust, Noah Bellwald, and Christoph Winkler

INTRODUCTION

Entrepreneurs constantly make decisions based on incomplete or inconsistent information due to high uncertainty in novel situations (Zander, 2007). Research has predominantly conceptualized learning in entrepreneurship contexts as a trial-and-error process that requires reflection on experience and successively leads to expertise in starting and growing new ventures (Mueller & Shepherd, 2016). However, entrepreneurs are often confronted with novel situations and problems where prior knowledge is not always applicable. Additionally, previous knowledge can even lead to bad decisions when behaviors are not adequately adapted to newly available information (Fust et al., 2018).

Successful entrepreneurs continuously evaluate whether their assumptions about current situations are still correct to adjust their strategies to environmental changes (Mishra, 2017). To successfully deal with uncertainty in novel situations, entrepreneurs need to learn beyond their pre-existing knowledge (Winkler et al., 2023). Despite its importance, only a few studies examine learning as a core capability of entrepreneurs (Solomon, 2007; Rauch et al., 2018). Given the importance of effective learning for success in entrepreneurship, we argue that it should be considered more than just a skill that develops with experience or as a by-product of project-based learning. Rather, we expand on current approaches with a learning-how-to-learn framework as a distinctive topic within entrepreneurship education scholarship.

Research on self-regulated learning (SRL) provides insights into how educators can enable students to take agency over their learning processes (Zimmerman, 2000, 2002). The concept of SRL goes beyond metacognition, as it also accounts for behavioral, social, and motivational influences of

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learning. SRL enables learners to plan, perform, and review their learning processes (Zimmerman, 1995). Thus, individual learning is performed in a more reflective and systematic way, resulting in higher learning capabilities in schools and sports (Bartulovic et al., 2017; Wilson et al., 2021). Studies in entrepreneurship argue that SRL enables entrepreneurs to learn in novel and uncertain situations (Fust et al., 2018; Winkler et al., 2023). Additionally, SRL helps to systematically develop entrepreneurial expertise, thus increasing the probability of success (Winkler et al., 2023; cf. Zimmerman, 2001). Research still lacks concrete insights about how educators can support nascent entrepreneurs to enhance their learning capabilities. To close this gap, we propose to answer the following research question: *How can entrepreneurship educators enable nascent entrepreneurs to improve their SRL capabilities?*

We contribute to entrepreneurship research mainly in two ways. First, we advance entrepreneurship education research by transferring and refining elaborated concepts of educational psychology. Our current understanding of entrepreneurship education is mainly based on experiential learning theories and focuses on venture development processes (Hägg & Gabrielsson, 2020). Scholars call for a more holistic perspective on learning processes within educational settings (cf. Hägg & Kurczewska, 2019; Hägg & Gabrielsson, 2020). Hence, we introduce the concept of self-regulated entrepreneurial learning (SREL) to foster higher-order learning in uncertain and dynamic contexts where prior experience can cause ineffective learning (Ucbasaran et al., 2011). Adapting Zimmerman's (2000, 2006) social cognitive model of SRL, Winkler et al. (2023, p. 2) describe SREL as a "cyclical process, wherein entrepreneurs plan and adapt their self-generated thoughts, feelings, and actions towards the attainment of personal entrepreneurial goals."

Second, researchers call for further work linking innovative learning theories with pedagogies to foster entrepreneurial learning (Hägg & Gabrielsson, 2020). Moreover, there seems to be a substantial research–practice gap in entrepreneurship education, which may hinder innovation within the landscape (Winkler, 2014). Pedagogies must, therefore, meet the innovative and creative mindset of nascent entrepreneurs (Solomon, 2007). We introduce a novel educational scaffold to foster SREL in entrepreneurship education and training programs (Bullough et al., 2015), testing its effectiveness in a real-life program.

THE BENEFITS OF SREL IN ENTREPRENEURSHIP EDUCATION

The teaching focus in entrepreneurship is still primarily on venture founding and development and not necessarily on higher-order thinking and learning processes when facing uncertainty. In contrast, many modern educational

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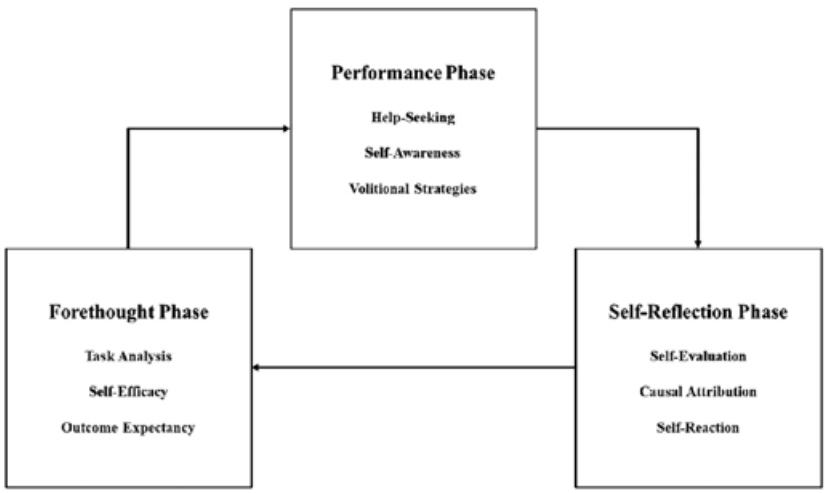
approaches such as discovery, problem- or inquiry-based learning follow the notion that learners should develop skills in planning, monitoring, and regulating their learning process in addition to the content matter (Thomas, 2009). We argue that SREL may give nascent entrepreneurs the means to improve their learning. By owning their cognition, behavior, and environment, they can develop and may better adapt their actions based on new information and environmental changes.

An adaptation of SRL (Zimmerman, 2001) to the entrepreneurship context was introduced by Fust et al. (2018). SREL enables learners to take ownership of their own learning process by systematically developing expertise to increase the probability of venture success (Figure 13.1). Educators could integrate SREL processes into frameworks such as entrepreneurship as a method (Neck & Greene, 2011) or the lean startup (Blank, 2013) to successfully approach challenges in opportunity identification and exploration. SREL consists of three phases: forethought, performance, and self-reflection (Zimmerman, 2002; Winkler et al., 2023).



Source: Adapted from Zimmermann & Campillo (2003).

Figure 13.1 SREL cycle

The forethought phase relates to the planning of the learning process and is about task analysis and self-motivational beliefs (Winkler et al., 2023). Task analysis consists of goal setting and the planning of learning activities (Locke & Latham, 2002). Highly self-regulated learners can break down long-term goals like venture growth or revenue into specific learning goals (Zimmerman,

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2007). Setting such intermediary process-oriented learning goals benefits self-improvement, especially in uncertain situations where new knowledge must be generated (Seijts & Latham, 2005). Entrepreneurs can gradually decrease their perceived uncertainty by identifying and planning the activities needed to obtain their previously set goals (McCann & Vroom, 2015). Key sources of self-motivational beliefs are self-efficacy and outcome expectancy (Zimmerman, 2012). Self-efficacy is the process where cognition, behaviors, and affects are systematically oriented toward attaining learning goals (Bandura, 1997). Learners with higher self-efficacy set higher goals and are more committed to achieving them (Locke & Latham, 2002). If self-efficacy is too high, entrepreneurs tend to overestimate their capabilities and set unreachable goals, which can decrease firm performance (Baron et al., 2016). Hence, the right calibration is important (Zimmerman, 2008). Outcome expectancy positively influences motivation as learners envision prospective rewards for their actions (Wigfield & Eccles, 2000). If the utility value of an upcoming task fits their strategic plans, the motivation to accomplish the task increases (Winkler et al., 2023).

The performance phase is about executing the planned (learning) activities and is concerned with metacognitive monitoring and self-control. Metacognitive monitoring enables learners to be self-aware about the situation, performance, and goal attainment (Zimmerman, 2006). This allows for corrective action if necessary. Additionally, self-regulated learners have a better awareness of their knowledge and limitations. These self-control mechanisms help to apply the right strategies like imagery, task and volitional strategies, environmental structuring, or help-seeking (Winkler et al., 2023; Zimmerman, 2002). Help-seeking, for example, is important to keep the passion for the venture over time and focus on one's core competencies (Collewaert et al., 2016).

The self-reflection phase concerns self-evaluation, causal attributions, and self-reactions (Winkler et al., 2023). Self-evaluation is the alignment of previously set goals and actual achievements. Entrepreneurs tend to set goals that are too challenging to achieve (Baron et al., 2016) or too far in the future (Winkler et al., 2023). By constantly comparing goals and achievements, entrepreneurs realize ambitious goals and learn to break down complex and long-term goals into smaller, attainable goals (Fust et al., 2018). More regular experiences of success increase self-efficacy and motivation (Zimmerman, 2007). The increased self-efficacy enables entrepreneurs to set more challenging but also more attainable goals in subsequent self-regulatory cycles (Winkler et al., 2023). Self-regulated learners who apply causal attribution are more likely to relate failure to poor strategy use than low ability (Zimmerman & Kitsantas, 1999). Consequently, they learn from this attribution more effectively (Ucbasaran et al., 2013). Self-reaction is the conclusion to what extent

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future actions need to be adapted to achieve desired goals (Winkler et al., 2023; Zimmerman, 2002), which has a huge impact on motivation, goal setting, and task strategies in subsequent self-regulatory learning cycles (Zimmerman, 2012).

To develop SREL capabilities of nascent entrepreneurs, learning environments and pedagogies in entrepreneurship classrooms should introduce the challenges of real-life entrepreneurship. Hence, entrepreneurship educators should foster higher-order learning skills such as SREL.

FOSTERING SREL

Experimental studies show that SREL can be scaffolded by interventions as it is a cyclical process consisting of three phases: forethought, performance, and self-reflection (Schunk & Ertmer, 2000). Each phase involves cognitive, motivational, and metacognitive processes (Zimmerman, 2000, 2006; Winkler et al., 2023). While an isolated training of subprocesses is not effective (Zimmerman, 1990), an instructional approach emphasizing metacognitive, cognitive, and motivational strategies is essential during SRL (Dignath et al., 2008).

The theoretical foundation of SREL is Bandura's (1977) social cognitive theory, which has concrete implications for the delivery of the training. Reciprocal determinism states that personal factors, including cognitive, emotional, and physiological aspects and environmental influences like observations of others, are reciprocally influencing each other. Hence, human beings are proactive and reflective in nature and influenced by their social environment (Pajares, 2012). Therefore, entrepreneurship educators should positively scaffold personal, behavioral, and environmental factors. Zimmerman (2001) theorizes that SREL capabilities develop across a four-level hierarchy: observation, emulation, self-control, and self-regulation.

To support observation, educators can provide participants with role models to learn from and detailed instructions on how to apply SREL (cf. Bandura, 1969). Being aware of this target behavior and how to get there enables the second development level, emulation (Zimmerman, 2001). During emulation external feedback is very important to improve self-reflection and self-awareness to align self-regulatory target behaviors (Zimmerman & Rosenthal, 1974). This initial guidance should be gradually reduced during SREL training. Participants can take over control of their learning processes and enhance SREL capabilities through independent practice (Ericsson et al., 1993). Thus, SREL is more than a teaching method where educators scaffold an individual's learning in context-specific learning situations. We therefore contend that SRL can also be applied to solving one's challenges in entrepreneurship. Self-control, the third level of SREL, refers to planning, executing,

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and monitoring the implementation of students' SREL strategies on their own (cf. Zimmerman, 2001). At this stage, learners are not necessarily reluctant to be in the presence of a role model anymore. Automating this behavior is the basis for the last and fourth level, self-regulation (Zimmerman, 2001). During this stage, learners can adapt their SREL strategies to changing individual and contextual conditions. They adjust behaviors based on their achievements. SREL is a cognitively and physically demanding process, and learners self-regulate their learning best if task value and self-efficacy beliefs are high (Zimmerman & Kitsantas, 1999).

ENTREPRENEURSHIP TRAINING

To design an entrepreneurship training intervention, Bullough et al. (2015) propose the entrepreneurship education and training (EET) effectiveness framework. Based on this framework, we designed a training scheme to promote SREL among nascent entrepreneurs, defined as individuals committed to creating a new venture by allocating a significant amount of their time to this endeavor. We emphasized training elements such as the background of the trainers and their role as coaches, the curriculum, and the delivery based on scaffolded SREL processes.

An ideal trainer background combines academic knowledge with hands-on experience in entrepreneurship to balance theory and action for participants (Pittaway & Cope, 2007). Trainers should enable learners to become managers of their own learning process. Therefore, it is important that they proactively facilitate individual or group work phases to discuss students' approaches and subsequently stimulate reflection (O'Donnell et al., 2002). Our SREL training curriculum consists of three pillars: (1) SREL instruction, (2) solving a challenge in entrepreneurship with the SREL canvas, and (3) motivational strategies and moderated group reflection. These pillars ensure that training elements align with training goals and the content applies to nascent entrepreneurs by providing early stage startup examples of role models to which the participants can relate to (Bullough et al., 2015). The delivery is based on scaffolded self-reflection exercises to increase participants' abilities to apply SREL gradually independently (cf. Yildiz & Celik, 2020; Zimmerman, 2001). The training takes approximately five hours.

SREL Instruction

The SREL instruction takes about one hour. The goal is to increase participants' awareness about the importance and benefits of SREL to address challenges in entrepreneurship like product development, pricing, or employee recruitment (Butler, 2002; Winkler et al., 2023. The SREL phases are explained with the

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help of role models, such as successful athletes or entrepreneurs, to whom learners can relate. Role models promote a positive transfer and increase achievement motivation (Bandura, 1997).

Entrepreneurship Challenge with the SREL Canvas

The second pillar takes approximately three hours. The goal for participants is to internalize cognitive, metacognitive, and motivational processes by completing a full SREL cycle with an individual current challenge in entrepreneurship. We specifically developed the SREL canvas (Figure 13.2) to facilitate this process by scaffolding learning activities in entrepreneurship. The SREL canvas maps the respective dimensions of entrepreneurial self-regulation according to Winkler et al. (2023), emphasizing forethought and self-reflection.

Trainers guide participants step by step through the canvas (Table 13.1). Inputs and exercises follow the four-level hierarchy of Zimmerman (2000): observation, emulation, self-control, and self-regulation. First, meaningful startup examples for observation are presented for each canvas step. Next, nascent entrepreneurs emulate patterns from the presented examples and apply the canvas to their own challenge to exert self-control. Finally, learners explain their approaches to their peers and discuss differences to enable the transfer of SREL strategies to new settings. This is repeated for all nine SREL canvas elements (Figure 13.2). Participants engage in joint reflections about their activities to promote learning. During this reflection, they pool their working memories to reduce cognitive load and increase metacognitive awareness (Sweller et al., 2019).

The goal of the forethought phase is to analyze the task at hand and activate self-motivational beliefs. Trainers enable participants to select an individual challenge, identify corresponding knowledge gaps, set learning, performance, and development goals, plan tasks, and determine respective self-efficacy levels (Table 13.1). An example of a relevant challenge could be the startup's product-market fit, where knowledge gaps may be related to lacking information about customer needs. Therefore, performance goals could be a segmentation of customers to determine appropriate target groups to be interviewed over the next three months. Corresponding learning goals would be to gather information about criteria for how the respective market could be segmented and a determination of the characteristics of the target group. Personal development goals bundle outcome expectations and increase the commitment toward set performance and learning goals (cf. Zimmerman, 2006, 2012). An example may be increasing one's competencies that can be applied in future market expansions to become a better entrepreneur.

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The objective of the performance phase is to stimulate metacognitive monitoring and self-control. Therefore, trainers guide participants to assess their required time for task completion, assess their target contributions, and evaluate their concentration. Learners discuss (potential) obstacles and the SREL strategies they applied in groups to foster learning and metacognitive awareness (Zimmerman, 2012). A typical obstacle, the planning fallacy, leads entrepreneurs to underestimate their task completion times by focusing only on future scenarios (Buehler et al., 1994). Especially in the founding phase of a startup, a wide range of activities (e.g., marketing, product development, finance, human resource management) is essential. To stay on track, entrepreneurs need to prioritize and share tasks.

The self-reflection phase intends to train self-evaluation and causal attributions. The training can only address one SREL cycle, so self-reaction can only be partially covered. Trainers scaffold participants to evaluate their performance, learning, and developing goals. This form of self-evaluative judgment is important to identify shortcomings with regard to goals set during the forethought phase. In addition, it builds the basis for forming causal attribution (cf. Zimmerman, 2001). To enable learning from failure and the adjustment of mental models and heuristics, trainers guide nascent entrepreneurs in attributing failure and success factors correctly (Ucbasaran et al., 2013). For instance, the focus could be on controllable variables such as planning time more generously for future tasks or interviewing more customers about their needs (cf. Zimmerman & Kitsantas, 1999).

Motivational Strategies and Moderated Group Reflection

Motivation is central to developing additional cognitive and metacognitive capabilities (Zimmerman & Campillo, 2003). Discussing motivational strategies and reflecting on the SREL cycle in our intervention takes approximately one hour. As motivation may vary during the challenge, the goal is to provide participants with both motivational and consolidated SREL strategies. According to Deci and Ryan's (2002) taxonomy of human motivation, several strategies, from addressing motivation to fostering intrinsic motivation, are provided, such as setting proximal goals, visualizing outcomes, developing routines, providing short-term rewards for finishing unpleasant tasks or restructuring the workplace. To foster SREL more generally, a combined perspective on cognitive, motivational, and metacognitive strategies is essential (Dignath et al., 2008). Hence, the trainer consolidates the different approaches applied during the workshop to enable learning from peers and to give a final holistic perspective on the SREL cycle.

The interaction of the different pillars is central to the successful implementation of our SREL training. First, SREL is introduced with theoretical

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Figure 13.2 SREL canvas

instruction. Second, SREL strategies are practiced with the support of an experienced trainer and the SREL canvas. In addition, prepared examples are shared to support the transfer of strategies to the task (self-reaction). Finally, motivational strategies are discussed and applied with SREL cycle reflection to finish the training.

QUANTITATIVE ASSESSMENT OF THE TRAINING

Study Design and Sample

Our study relies on a pre-/post-test control group design (cf. Rauch & Hulsink, 2015). We applied a quasi-experimental approach since we were not able to randomly assign nascent entrepreneurs to the treatment condition and the control groups (Cook et al., 1990). We ran training workshops with two control groups, to increase power and test for potential trainer effects. The treatment group (TG) consisted of 37, and both control groups (CG 1 & CG 2) of 27 participants (Table 13.2). Both control groups received entrepreneurship training without explicit SREL scaffolds. The curriculum and the delivery of the three workshops offered an adequate comparative basis for the purposes of this study. Both treatment and control groups participated in a pre-test (T1) prior to the training and a post-test (T2) immediately following the training.

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Effects on SREL Capabilities

To measure SREL, we applied three distinct scales, each specifically designed to assess one SREL phase within the context of entrepreneurship (Bellwald et al., 2023).

A paired t-test indicated a substantial increase in post-intervention ($t(31) = 6.40$, $p < .001$), with mean scores rising from $M = 3.80$ ($SD = 0.43$) pre-intervention to $M = 4.25$ ($SD = 0.46$) post-intervention in TG (Table 13.3). This change represents a large effect size ($r = 1.13$) according to Connelly et al. (2010). Conversely, the control groups did not exhibit significant changes in their SREL scores, with minimal fluctuations from pre-test (CG1: $M = 3.81$, $SD = 0.24$; CG2: $M = 3.76$, $SD = 0.23$) to post-test (CG1: $M = 3.81$, $SD = 0.24$; CG2: $M = 3.76$, $SD = 0.23$). Comparisons between TG und CGs highlighted

Table 13.1 Overview of the entrepreneurship training

Curriculum		Delivery	Trainer	Training goals	Success factors
SREL Instruction					
Introduction		Welcoming participants, moderating the introduction round, and asking participants about their motivation.	A	Overview of participants	Open-minded atmosphere
Entrepreneurship, SRL and SREL		Explaining reasons for startup failures and discussing the meaning of entrepreneurship.	A	Relevance of SREL	
		Introducing the SRL cycle, the rationale behind it, and its benefits for entrepreneurs. Transfer from SRL in schools to SREL in entrepreneurship. Presentation of successful athletes and entrepreneurs and their SRL approaches to challenges.	A	Benefits and overview of the SRL cycle	Role models that can be identified with
Entrepreneurship Challenge with the SREL Canvas					
SREL canvas		Introducing the SREL canvas and explaining the three phases of forethought, performance, and self-reflection with examples. Introducing a prepared example startup, which will be referred to during the complete SREL cycle.	E	Relevance and overview of the SREL cycle	Startup that can be related to
Forethought phase	Entrepreneurship challenges	Explaining typical challenges in entrepreneurship. Presenting the selected challenge of the example startup. Guiding participants in identifying their own challenge to be addressed during the training. Moderating group discussion concerning the identified challenges.	E	Task analysis: identifying relevant challenges	Challenges are urgent and of high relevance
	Knowledge gaps	Explaining how to identify knowledge gaps with the help of the Rumsfeld matrix. Introducing the concept of situational awareness as the perception and understanding of critical factors in the business environment. Presenting the knowledge gaps of the startup example. Guiding participants in reflecting on their situational awareness and identifying their own knowledge gaps concerning their challenges. Moderating group discussion concerning identified gaps.	E	Task analysis: identifying knowledge gaps	Situational awareness is understood and can be applied
	Goal setting (performance goals, learning goals, and development goals)	Explaining performance goals and the SMART principle (specific, measurable, assignable, realistic, and time-related). Presenting the performance goals of the startup example. Guiding participants in formulating SMART performance goals. Moderating group discussion concerning set goals.	E	Task analysis: setting performance goals	Connection between challenges, knowledge gaps, performance goals, learning, goals, and development goals is understood
		Explaining learning goals, their connection to performance goals, and their importance to learning in novel situations. Presenting learning goals of the startup example. Guiding participants in deviating their own learning goals from performance goals. Moderating group discussion concerning set goals.	E	Task analysis: setting learning goals	
		Explaining development goals and their positive effect on the commitment towards performance and learning goals. Presenting development goals of the startup example. Guiding participants in thinking about and defining their own development goals. Moderating group discussion concerning set goals and thoughts.	E	Outcome expectancy: reflecting development goals	
	Task analysis and strategic planning	Introducing task definition, prioritization, formulation of approaches, and target times. Explaining how to define tasks out of learning goals. Presenting tasks and strategic planning of the startup example. Guiding participants in defining their own tasks, finding approaches, estimating target times, and prioritizing. Moderating group discussion concerning planning activities.	E	Task analysis: breaking goals into tasks and learning how to plan and prioritize	Planning fallacy and overconfidence bias are incorporated
	Self-efficacy	Explaining self-efficacy and providing negative examples of entrepreneurs with too low self-efficacy beliefs and too high self-efficacy beliefs. Sensitizing that being overconfident can mislead to ignoring new or contradictory information. Being anxious may lead to procrastination. Guiding participants to assess their self-efficacy for the defined tasks. Moderating group discussion concerning calibration strategies and providing new strategies for the regulation of self-efficacy.	E	Self-efficacy: Reflecting and calibrating self-efficacy	

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Performance phase	Action	Guiding participants in selecting a minor task like market research or expert calls and providing them with some time to perform it. Stimulating reflection of own limitations, where to find the needed information to address set goals, and how to deal with obstacles.	E	Help-seeking: finding the right information	Prioritization and task sharing are applied
	Time management, target contribution, and concentration	Explaining to participants how to monitor activities and timing. Sensitizing that being self-aware about own performance and goal attainment allows for corrective actions where needed. Presenting the startup example's comparison of target and needed time, target contributions of activities, and learnings. Moderating group discussion about self-monitoring activities and learnings	E	Self-awareness: Observing the progress of activities	Monitoring and working behavior examples are comprehensible and can be related to
		Explaining to participants how to monitor and improve their working behavior. Encouraging learners to experiment with SREL strategies to identify the most productive working conditions. Presenting monitoring activities and working behavior improvements of the startup example. Moderating group discussion about reflections of own working preferences.	E	Volitional strategies: regulating working behavior	
Self-reflection phase	Comparison of goals and results	Explaining how to compare goals and results. Presenting the startup example's comparison. Providing only a little time as the performance phase was covered partly only, too.	E	Self-evaluation: evaluating failure and success	Keeping it short
	Causal attribution	Explaining causal attribution and the attribution bias. Introducing negative examples of entrepreneurs suffering from attribution bias. Presenting causal attribution notes of the startup example. Guiding participants in attributing success and failure to internal and external influences. This is not restricted to the previous performance phase; also, past challenges can be reflected upon. Moderating group discussion about causal attributions.	E	Causal attribution: attributing failure and success	Active support of trainers that participants learn to attribute correctly
	Key learnings	Explaining how to identify key learnings. Presenting key learnings of the startup example. Guiding participants in identifying their key learnings. Unfortunately, self-reaction cannot be transferred into the forethought phase of the next challenge. Moderating reflections on key learning results for future challenges.	E	Self-reaction: identifying key learnings for future challenges	Establish the connection to the forethought phase of the next challenge
Motivational Strategies and Moderated Group Reflection					
Motivational strategies		Explaining the taxonomy of human motivation and providing examples for each dimension. Guiding participants to classify their motivation for a self-selected task according to the taxonomy. Supporting to reflect own motivational strategies. Moderating of group discussions about motivational strategies.	A	Volitional strategies: Identifying sources of motivation and resilience	Promote learning from peers
SREL strategies		Consolidating SREL strategies presented by trainers and applied by participants. Moderating group discussion about applied SREL strategies and learnings in general. Giving a final holistic perspective on SREL combining cognitive, metacognitive, and motivational strategies.	A	Overview of the SREL cycle and SREL strategies	Combining cognitive, metacognitive, and motivational strategies

Note: E = Entrepreneur; A = Academic.

Table 13.2 Sample characteristics

		TG		CG 1		CG2	
		N	%	N	%	N	%
Gender	Male	19	59	9	82	10	62.5
	Female	13	41	2	18	6	37.5
Age		32	23	11	25	16	25
Entrepreneurial background parents	Yes	20	62.5	6	55.5	7	44
	No	12	37.5	5	45.5	9	56
Entrepreneurial intention	Yes	30	94	6	54.5	15	94
	No	2	6	5	45.5	1	6
SREL baseline	M	32	3.80	11	3.81	16	3.76
	SD		.43		.23		.23

a significantly greater improvement in the intervention group, underscoring the effectiveness of the SREL-focused intervention.

Our analysis commenced with a Shapiro-Wilk test, confirming that CG1's and CG2's SREL scores were normally distributed, unlike those of the TG.

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Table 13.3 Paired t-test: means, significance, and effect size

			Pre-test (T1)		Post-test (T2)					
Variable	Group	N	M	SD	M	SD	T	df	p	r
SREL	TG	32	3.80	.43	4.25	.46	6.4	31	.000	1.131
	CG1	11	3.81	.23	3.74	.26	1.12	10	.289	
	CG2	16	3.76	.23	3.70	.29	1.11	15	.285	

This necessitated further non-parametric testing. The Kruskal-Wallis test indicated no significant differences across groups' pre-intervention ($H = 0.284$, $p = 0.868$), suggesting comparable baseline SREL capabilities. However, post-intervention, significant differences emerged ($H = 19.065$, $p < .001$), substantiating the intervention's impact. Dunn's post hoc comparisons revealed marked SREL improvements in TG when juxtaposed with CG1 and CG2 ($p < .01$ and $p < .001$, respectively), validating the workshop's effectiveness in enhancing self-regulatory efficacy.

DISCUSSION

This chapter aimed to design an entrepreneurship training intervention to support the development of entrepreneurs' SREL capabilities, which are essential to developing entrepreneurial expertise (Fust et al., 2018; Winkler et al., 2023). To date, there has been little empirical evidence of how SRL skills can be scaffolded within an entrepreneurship education context. Building on the reciprocal relationship of environment, cognition, and behavior, we want to close the gap between entrepreneurship research and practice (Winkler, 2014). Our study demonstrated that participants significantly improved their SREL capabilities. The control groups did not improve their SREL capabilities, despite having had a similar interactive workshop dealing with venture development.

We contribute to the entrepreneurship education literature in two ways. First, we introduced SREL as an essential element in entrepreneurship education. Entrepreneurs with distinct SREL capabilities develop and adapt their strategies based on new information or environmental changes. By taking agency over their cognition, behavior, and the environment nascent entrepreneurs can learn beyond their pre-existing knowledge and are subsequently better able to deal with uncertainty in novel situations (Fust et al., 2018; Winkler et al., 2023). Integrating SREL scaffolds into educational training programs enables drawing explicit linkages to potential knowledge gaps, which is essential for learning in uncertain and dynamic environments (Fust et al., 2018).

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Second, we developed and tested a replicable training model that entrepreneurship educators can easily adapt. Our study showed a significant increase in SREL capabilities when scaffolding entrepreneurs' cognitive, motivational, and metacognitive processes during an entrepreneurship training intervention, supporting Dignath et al.'s (2008) findings. Therefore, we identified three essential training elements: curriculum, delivery, and trainers' roles (Bullough et al., 2015).

Our SREL canvas can support the delivery as a learning scaffold. It should be used to tackle a challenge in entrepreneurship based on specific target group examples and scaffolded based on the four development levels, according to Zimmerman (2001). We examined how an individual challenge increases the relevance and engagement of participants.

The focus of the curriculum should be on the cyclical three-phase SREL model. Nevertheless, an introduction where the rationale and relevance of SREL are underlined is important to further catch participants' attention. Moreover, forethought and self-reflection both strongly relate to the performance phase based on planning activities and self-reflections. During the forethought phase, the emphasis is on sequentially breaking one's complex challenge into manageable tasks and defining suitable approaches (Table 13.1). The reflection phase should include previous problem-solving activities to promote metacognitive and motivational SREL strategies. It is important to continuously draw the connection between the SREL strategies of the different phases as they build on each other and are reciprocal in nature.

The SREL canvas allows nascent entrepreneurs to address a challenge of their own and take control of their learning process. Feedback from trainers and peers improves self-reflection and self-awareness (Zimmerman, 2001; O'Donnell et al., 2002). This guidance should be gradually reduced during the training while increasing self-reflection. This way, learning can be stimulated even further and evolve to higher SREL levels (cf. Rogers, 2001; Zimmerman, 2001). Subsequent group discussions help to increase peer learning.

We conclude that the scaffolding-oriented delivery of the SREL canvas provides interested entrepreneurship educators with a blueprint for an entrepreneurship training intervention to support SREL among nascent entrepreneurs.

LIMITATIONS AND FUTURE RESEARCH

The study has several limitations that future research could address. First, we applied a quasi-experimental study design and could not randomly assign participants to the treatment and control conditions. Therefore, we were not able to control for initial group differences and infer causality. Moreover, the self-selection could bias results in favor of our entrepreneurship training and affect generalizability. Second, due to the limited availability of nascent

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entrepreneurs, we were not able to conduct a follow-up survey and test for long-term effects. Third, the small sample size may result in low power of the study. However, small samples are not unusual in field experiments to reduce the complexity (Eden, 2017).

Future research could address these shortcomings by designing a true experiment with a bigger sample size where nascent entrepreneurs are randomly assigned to treatment and control groups. In addition, we recommend conducting a post-survey several months after the intervention to evaluate the long-term effects of the SREL intervention.

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